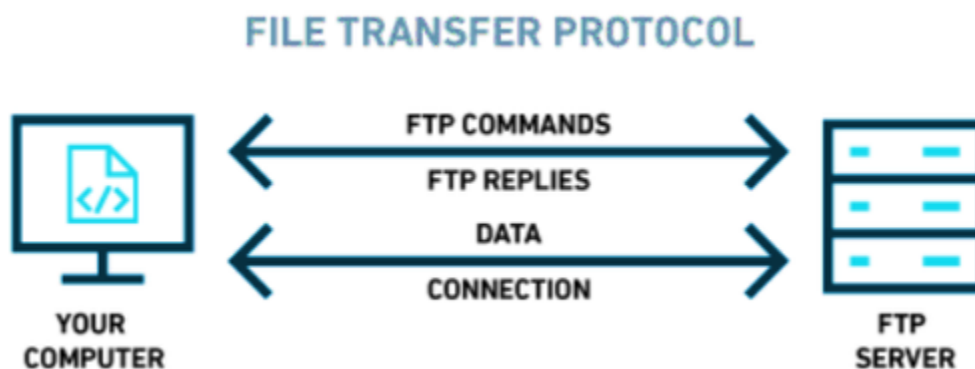


TFTP (Trivial File Transfer Protocol):

- o TFTP is term, which is stand for **Trivial File Transfer Protocol**.
- o Trivial File Transfer Protocol is the simpler file transfer protocol.
- o TFTP is used to save and restore a Router or Firewall configuration.
- o TFTP is used to backup, restore and upgrade IOS image of devices.
- o Trivial File Transfer Protocol uses UDP (User Datagram Protocol).
- o Trivial File Transfer Protocol uses a well-known UDP port number 69.
- o Trivial File Transfer Protocol requires very small amount of memory.
- o TFTP uses connection-less User Datagram protocol for communication.
- o Trivial File Transfer Protocol (TFTP) is a simple file transfer protocol.
- o TFTP is insecure protocol because it don't have Authentication mechanism.
- o TFTP does not support user authentication and all data is sent in clear text.
- o Trivial File Transfer Protocol uses less resource as compare to FTP protocol.
- o Trivial File Transfer Protocol doesn't provide user authentication mechanism.

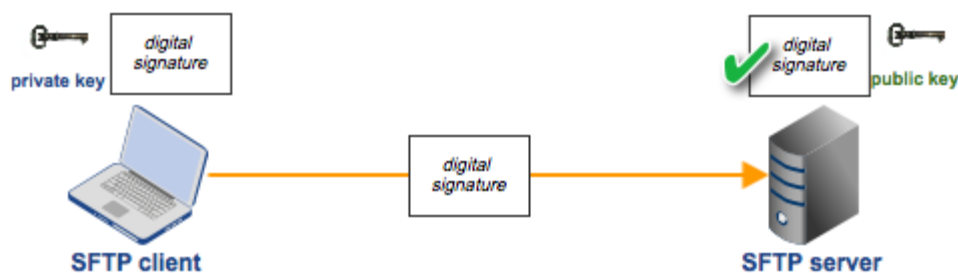
FTP (File Transfer Protocol):

- o FTP is Network Term, which is stands for **File Transfer Protocol**.
- o FTP is used to send and receive file from the remote device Routers.
- o File Transfer Protocol is a complete, session-oriented file transfer protocol.
- o FTP depends on TCP, is connection oriented, and provides reliable control.
- o File Transfer Protocol provides user authentication mechanism & method.
- o File Transfer Protocol (FTP) uses usernames and passwords for setup.
- o Devices are required to have a username and password setup for FTP.
- o File Transfer Protocol (FTP) is faster when compared to TFTP Protocol.
- o FTP uses TCP ports 20 for sending data & 21 for sending control commands.
- o FTP supports Authentication, but there is no Encryption in FTP transfer.



SFTP:

- o SFTP is network term, which is stands for **Secure File Transfer Protocol**.
- o SFTP is preferred over FTP protocol whose operation is performed in plain text.
- o SSH, File Transfer Protocol or SFTP is a network protocol that provides file transfer.
- o SFTP uses Secure Shell (SSH) in its underlying process to securely transfer the data.
- o Secure File Transfer Protocol SFTP has other capabilities beside secure file transfer.
- o SFTP clients resume interrupted file transfers, removing specific files & directory listings.
- o SSH includes support for SFTP new standard file transfer protocol introduced in SSHv2.
- o This feature provides a secure and authenticated method for copying device configuration.
- o SFTP, this feature also provides a secure method and way for copying device image files.
- o SFTP must require enabling SSH & ip SSH source-interface command must be configured.

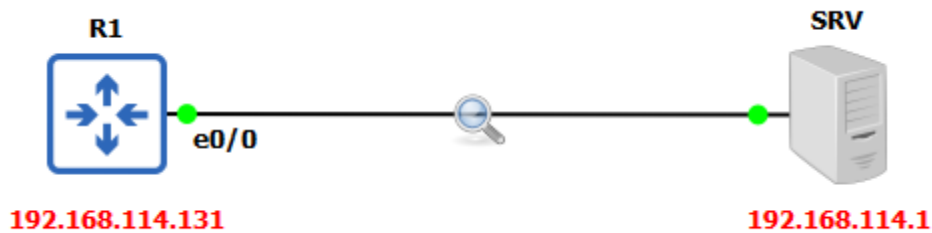


SCP:

- o SCP is network term, which is stands for **Secure Copy** is a File Transfer Protocol.
- o SCP provides secure & authenticated method for copying device configurations.
- o SCP also provides secure & authenticated method for copying device image files.
- o Secure Copy Protocol relies on SSH, before enabling SCP, correctly configure SSH.
- o Also, enable authentication and authorization on the Router and Switch for SCP.
- o SCP (Secure Copy) runs over TCP port number 22 by default which is port for SSH.
- o SCP (Secure Copy Protocol) is also a Connection Oriented protocol secure like SFTP.
- o It is used for sensitive & rapid file transfer & Authentication and Encryption is used.



Lab Time:



R1 Basic Configuration

```
R1(config)#interface e0/0
R1(config-if)#ip address dhcp
R1(config-if)#no shutdown
```

FTP Configuration

```
R1(config)#ip ftp username admin
R1(config)#ip ftp password 123456

R1#copy running-config ftp:
Address or name of remote host []? 192.168.114.1
Destination filename [r1-confg]?
```

Wireshark Capture FTP sending all the data and details in clear text.

The image shows a Wireshark packet capture of an FTP session. The title bar indicates the capture is on the interface 'SRV nat0 to R1 Ethernet0/0'. The packet list shows several FTP packets. The 'Info' column for packet 66 shows 'Request: USER admin' with 'USER admin' highlighted in a red box. Packet 67 shows 'Request: PASS 123456' with 'PASS 123456' highlighted in a red box. Packet 69 shows 'Request: STOR r1-confg'. The packet details pane is currently empty.

Time	Source	Destination	Protocol	Length	Info
38.962971	192.168.114.1	192.168.114.131	FTP	96	Response: 220 3Com 3C Daemon FTP Server Version 2.
38.963799	192.168.114.131	192.168.114.1	FTP	66	Request: USER admin
39.069370	192.168.114.1	192.168.114.131	FTP	87	Response: 331 User name ok, need password
39.070033	192.168.114.131	192.168.114.1	FTP	67	Request: PASS 123456
39.072017	192.168.114.1	192.168.114.131	FTP	74	Response: 230 User logged in
39.072678	192.168.114.131	192.168.114.1	FTP	62	Request: TYPE I
39.074919	192.168.114.1	192.168.114.131	FTP	74	Response: 200 Type set to I.
39.075536	192.168.114.131	192.168.114.1	FTP	60	Request: PASV
39.077866	192.168.114.1	192.168.114.131	FTP	105	Response: 227 Entering passive mode (192,168,114,
39.079609	192.168.114.131	192.168.114.1	FTP	69	Request: STOR r1-confg
39.082138	192.168.114.1	192.168.114.131	FTP	98	Response: 125 Using existing data connection

TFTP Configuration

```
R1#copy running-config tftp:
Address or name of remote host []? 192.168.114.1
Destination filename [r1-confg] R1-CFG
```

SCP Configuration

```
R1(config)#ip domain-name test.com
```

```
R1(config)#crypto key generate rsa modulus 1024
```

```
R1(config)#username admin privilege 15 password 123
```

```
R1(config)#ip scp server enable
```

```
R1(config)#line vty 0 4
```

```
R1(config-line)#transport input ssh
```

```
R1(config-line)#login local
```

```
R1(config)#ip scp server enable
```

```
c:\> pscp.exe -scp admin@192.168.114.131:running-config d:\abc.txt
```

```
c:\> pscp.exe -scp admin@192.168.114.131: startup-config d:\startup-config.txt
```

```
C:\> pscp.exe -scp admin@192.168.114.131:running-config d:\abc.txt
The server's host key is not cached in the registry. You
have no guarantee that the server is the computer you
think it is.
The server's rsa2 key fingerprint is:
ssh-rsa 1024 b9:78:44:b4:4b:81:6d:b3:8c:9d:a2:f1:3d:32:6c:8c
If you trust this host, enter "y" to add the key to
PuTTY's cache and carry on connecting.
If you want to carry on connecting just once, without
adding the key to the cache, enter "n".
If you do not trust this host, press Return to abandon the
connection.
Store key in cache? (y/n) y
Keyboard-interactive authentication prompts from server:
| Password:
End of keyboard-interactive prompts from server
abc.txt | 1 kB | 1.4 kB/s | ETA: 00:00:00 | 100%
```

*- [SRV nat0 to R1 Ethernet0/0]

Time	Source	Destination	Protocol	Length	Info
8699.142774	192.168.114.131	192.168.114.1	SSHv2	106	Server: Encrypted packet
8699.143108	192.168.114.1	192.168.114.131	SSHv2	242	Client: Encrypted packet
8699.144109	192.168.114.131	192.168.114.1	SSHv2	90	Server: Encrypted packet
8699.144515	192.168.114.1	192.168.114.131	SSHv2	142	Client: Encrypted packet
8699.246014	192.168.114.131	192.168.114.1	SSHv2	122	Server: Encrypted packet
8699.246583	192.168.114.1	192.168.114.131	SSHv2	142	Client: Encrypted packet
8699.247687	192.168.114.131	192.168.114.1	SSHv2	614	Server: Encrypted packet
8699.247814	192.168.114.131	192.168.114.1	SSHv2	614	Server: Encrypted packet
8699.247915	192.168.114.131	192.168.114.1	SSHv2	426	Server: Encrypted packet
8699.254916	192.168.114.1	192.168.114.131	SSHv2	142	Client: Encrypted packet